Code Roulette: How Prompt Variability Affects LLM Code Generation

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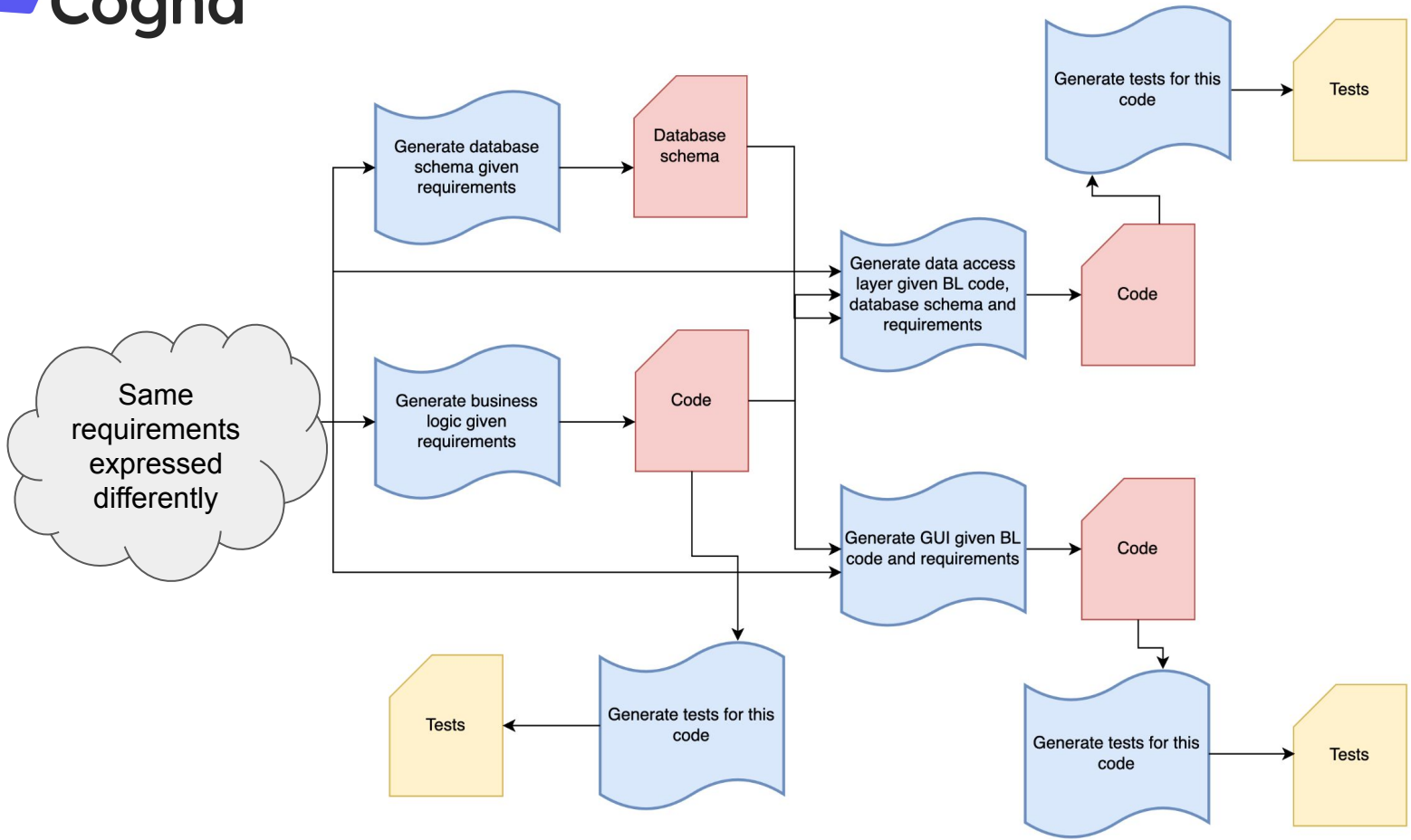
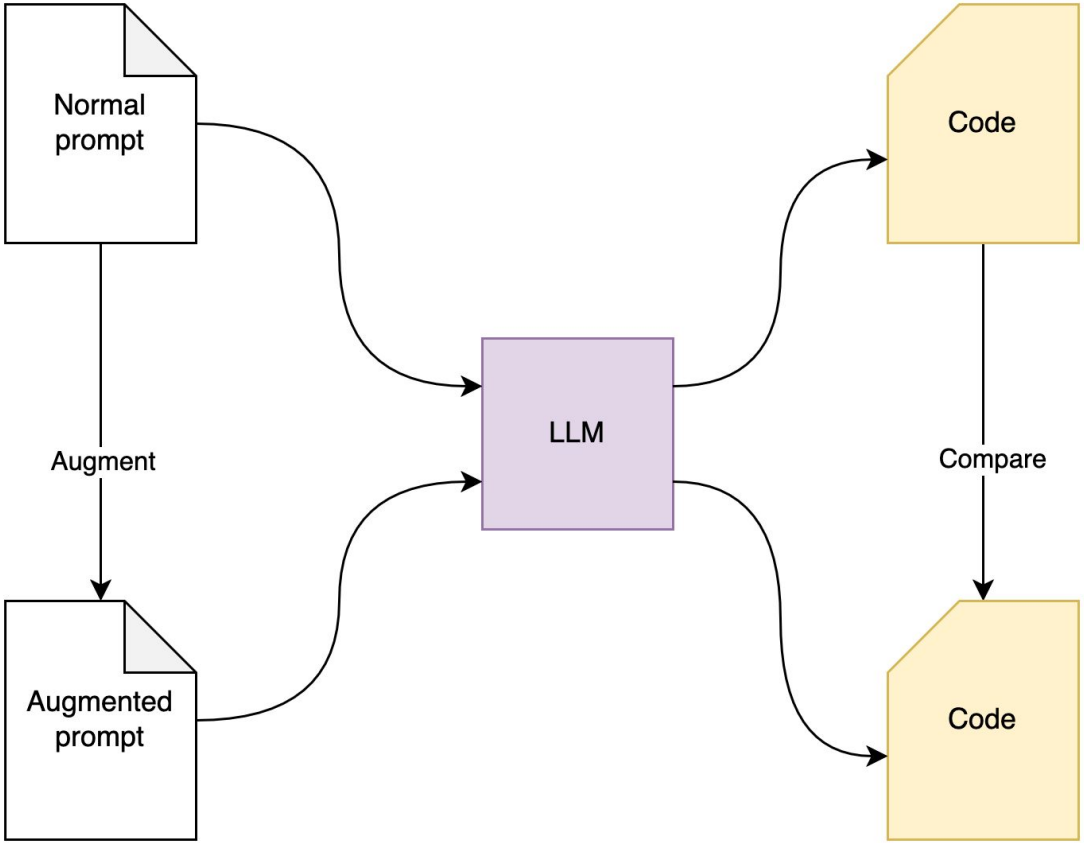
LLM4Code workshop @ ICSE 2026

Cogna


Same requirements expressed differently
Generate database schema given requirements
Database schema
Generate business logic given requirements
Code
Generate data access layer given BL code, database schema and requirements
Code
Generate tests for this code
Tests
Generate GUI given BL code and requirements
Code
Generate tests for this code
Tests
Generate tests for this code
Tests

RQ: How sensitive is LLM code generation to variations in requirements?

Overview of evaluation



Augment

a → s

1. Typos

speak → talk

2. Synonyms

Write a program → Develop an app

3. Paraphrasing

Edward Ma, NLPaug, https://github.com/makcedward/nlpaug

LLM

Gemini 2.0

Claude 3

GPT-4

Llama 3.3

Compare

❌ Sacre BLEU

❌ Jaccard similarity

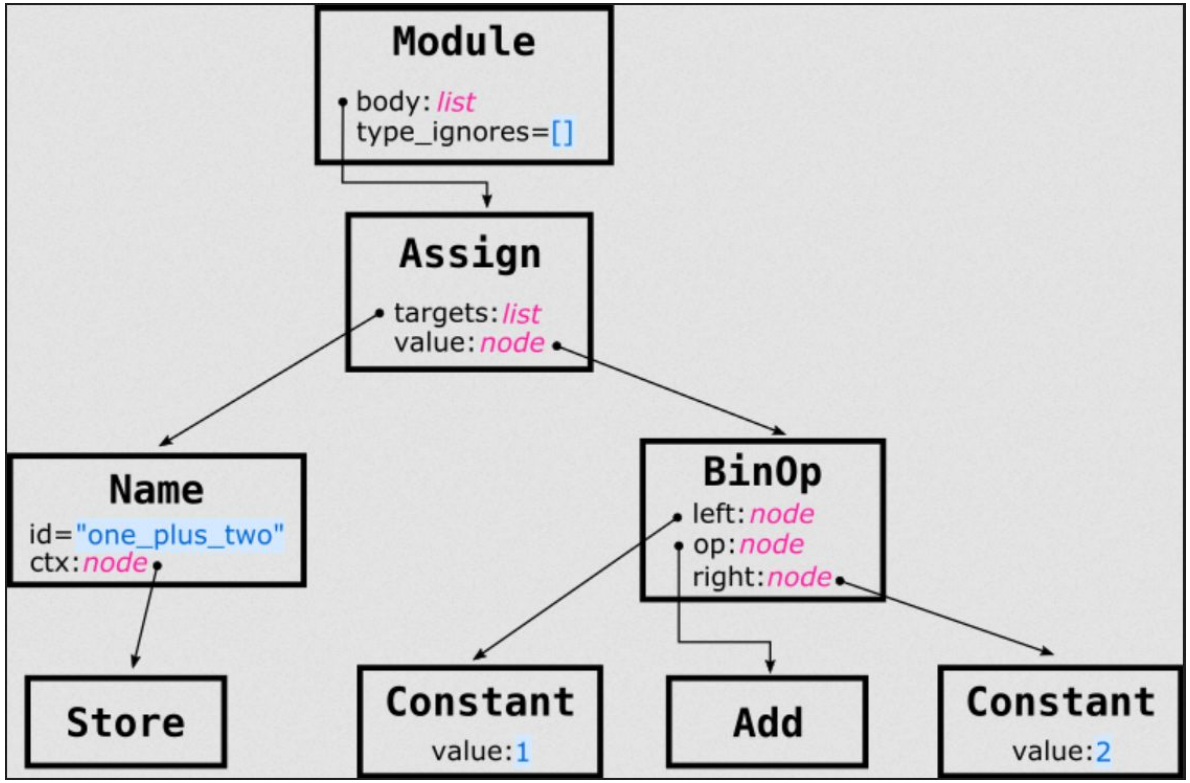
❌ BERT score

✔ Tree Similarity of Edit Distance (TSED)

Evtikhiev et al., “Out of the Bleu: how should we assess quality of the code generation models?”, Journal of Systems and Software, 2023
Pawlik and Augsten, “Efficient computation of the tree edit distance”. ACM Transactions on Database Systems (TODS), 2015
Song et al., “Revisiting Code Similarity Evaluation with Abstract Syntax Tree Edit Distance”, ACL 2024

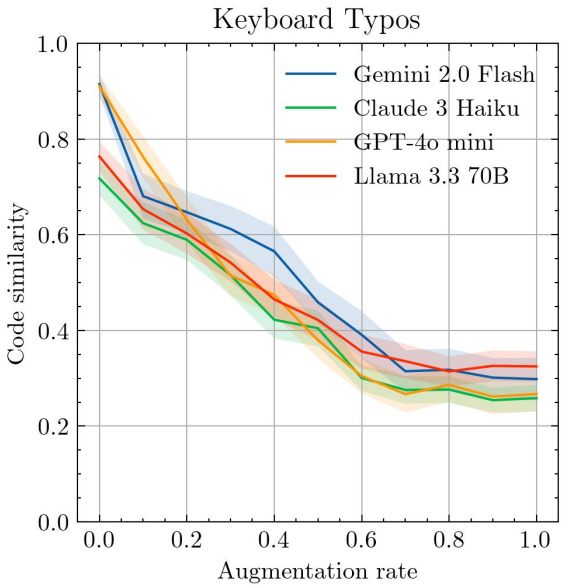
Abstract Syntax Tree

```
one_plus_two = 1+2
```



Alessandro Finamore, “Abstract Syntax Trees in Python”, https://pybit.es/articles/ast-intro/

Results: typos example



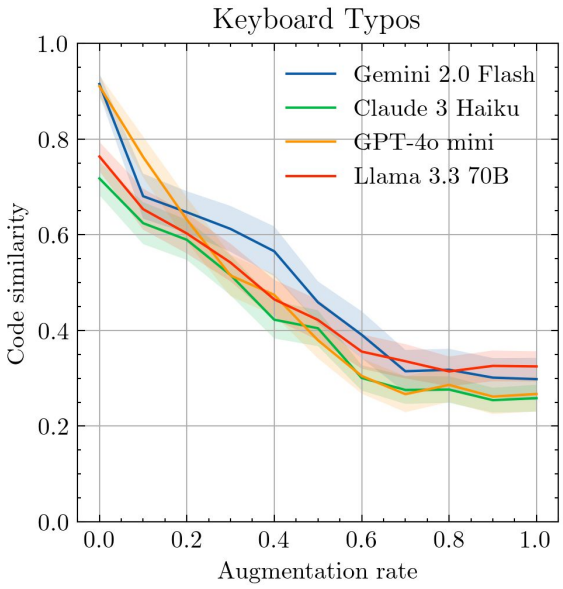
Rate = 0

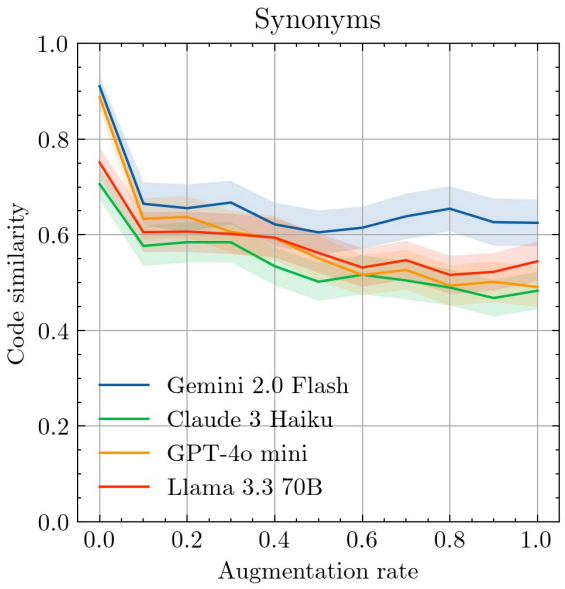
Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to target.

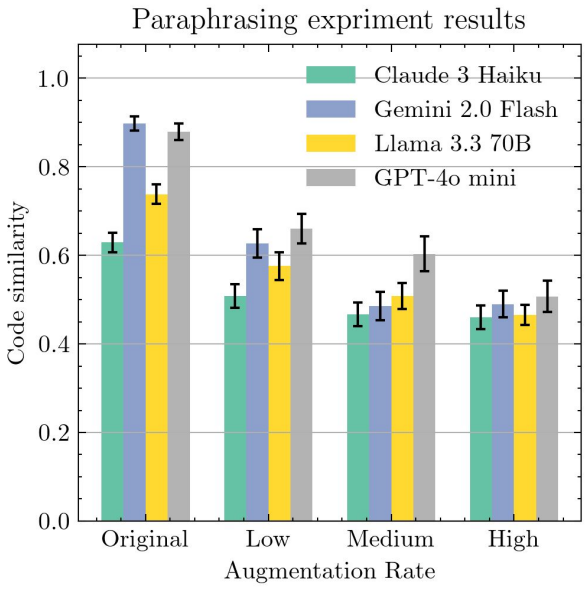
Rate = 1

NLGWj an x%5W^ of KJH#VWFx H&Jz and an 8jRdb#T ^sTHWF, tS4Kgg 7MRlssz of the two b T<N@fe Z tDB GbSf fyD% add up to rW5v4G.

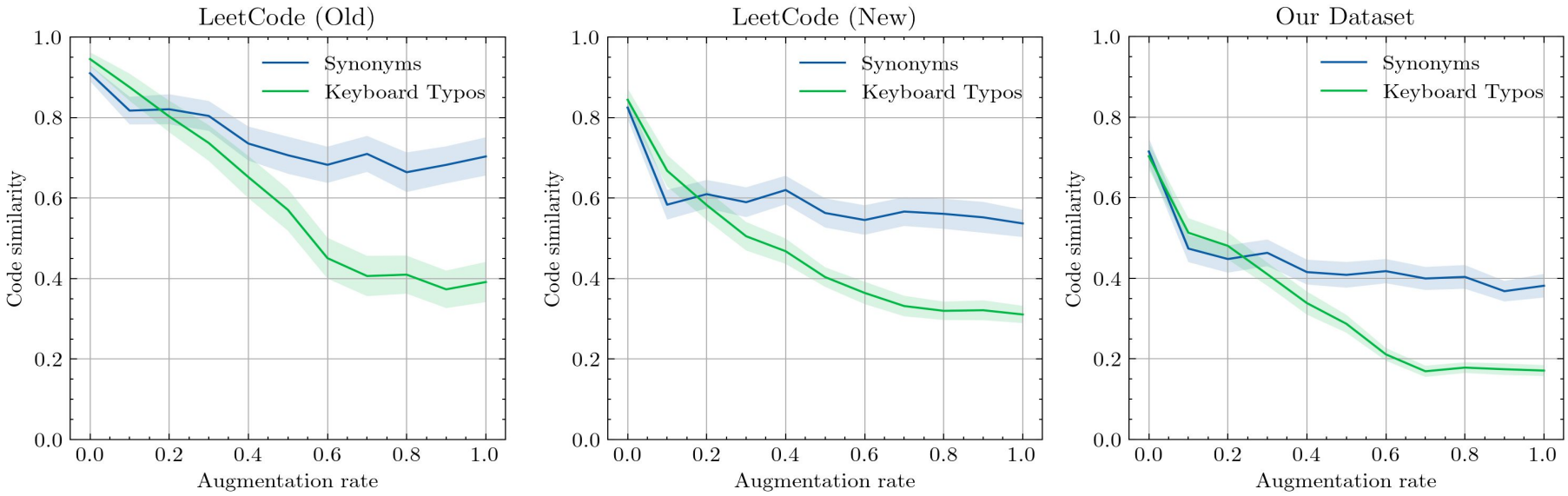
Results: overall







Results: sensitivity to dataset recency



LLMs are somewhat sensitive. So what?

- We describe full procedure to measure sensitivity
- Behavior is nuanced between LLMs
- Input data matters!...
- …as well as type of augmentation

Questions?

https://mlatcl.github.io/

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